

BST Casimir Stability: Seeley-DeWitt Analysis

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Summary

This note reports the results of a rigorous Seeley-DeWitt analysis of the Casimir spectral zeta function $\zeta_{S^4 \times S^1}(-1/2; \rho)$ with $\rho = R_s/R_b$. The goal was to test the Casimir Stability Conjecture (Section 5.3 of WorkingPaper.md): does this zeta function have a minimum at $\rho_0 = 137$?

Short answer: The standard Casimir mechanism does NOT produce a minimum at $\rho = 137$. However, the Wyler formula gives $\alpha = 1/137.036$ directly from the D_{IV}^5 geometry with 0.0001% precision. The two results are complementary: Wyler derives the value; Casimir stability would confirm it is dynamically stable.

1. What Was Computed

Using the Poisson resummation of the S^1 modes, the spectral zeta function splits as:

$$\zeta(-1/2, \rho) = \underbrace{\zeta^{UV}(-1/2, \rho)}_{\text{UV piece}} + \underbrace{\zeta^{\text{fin}}(-1/2, \rho)}_{\text{finite piece}}$$

1.1 The UV Piece

The $n = 0$ Poisson term gives:

$$\zeta^{UV}(s, \rho) = \rho \sqrt{\pi} \cdot \frac{\Gamma(s - 1/2)}{\Gamma(s)} \cdot \zeta_{S^4}(s - 1/2)$$

This is proportional to ρ (exact at all orders). At $s = -1/2$, $\Gamma(s - 1/2)$ has a simple pole. Extracting the finite part (zeta regularization):

$$\zeta^{UV, \text{ren}}(-1/2, \rho) = \frac{1 - \gamma_E}{2} \cdot \zeta_{S^4}(-1) \cdot \rho = C_{UV} \cdot \rho$$

where $\gamma_E = 0.5772\dots$ is the Euler-Mascheroni constant and $\zeta_{S^4}(-1)$ is the analytically continued S^4 spectral zeta.

Numerical result:

$$\zeta_{S^4}(-1) = 0.015170 \quad (\text{Hurwitz/Bernoulli method})$$

$$C_{UV} = \frac{1 - \gamma_E}{2} \times 0.015170 = +0.003207$$

Since $C_{UV} > 0$, the UV piece is monotonically increasing in ρ — no extremum.

1.2 The Finite Piece (Winding Modes)

The $n \geq 1$ Poisson terms give:

$$\zeta^{\text{fin}}(-1/2, \rho) = -\rho \sum_{n=1}^{\infty} I_n(\rho)$$

where:

$$I_n(\rho) = \int_0^{\infty} dt t^{-2} K_{S^4}(t) e^{-\pi^2 n^2 \rho^2 / t}$$

Saddle point analysis: At $\rho = 137$, the saddle is at $t^* = \pi n \rho / 2 \approx 215$ (for $n = 1$). At this value, $K_{S^4}(t^*) \sim e^{-4 \times 215} = e^{-860}$. The dominant exponential factor is:

$$I_n(137) \sim e^{-4\pi n \times 137} = e^{-1722n}$$

For $n = 1$: $I_1(137) \sim 10^{-748}$. Numerically zero. [Corrected 2026-10-07, Elie toy 5862: this is the $l \geq 1$ part only. With the $l = 0$ zero mode, which the code includes, $I_1(137) \approx 1/(\pi^2 \rho^2) \approx 5.4 \times 10^{-6}$, and the winding piece is the circle Casimir term $-1/(6\rho)$.]

The finite Casimir piece is only significant for $\rho \lesssim 5$:

ρ	$I_1(\rho)$	ζ^{fin}
0.5	4.19×10^{-1}	-0.303
1.0	1.01×10^{-1}	-0.148
2.0	2.53×10^{-2}	-0.074
5.0	4.05×10^{-3}	-0.030
137	$\sim 10^{-748}$	≈ 0

1.3 Total Renormalized Zeta

$$\zeta^{\text{ren}}(-1/2, \rho) = 0.003207 \rho - \rho \sum_{n=1}^{\infty} I_n(\rho)$$

This function is monotonically increasing for $\rho \gtrsim 5$ (UV piece dominates). There is no minimum at $\rho = 137$.

2. The Wyler Formula — the Correct Derivation of $\rho = 137$

While the Casimir mechanism does not select $\rho = 137$, the Wyler formula does so with geometric precision:

$$\alpha = \frac{9}{8\pi^4} \left(\frac{\pi^5}{2^4 \cdot 5!} \right)^{1/4} = \frac{9}{8\pi^4} \left(\frac{\pi^5}{1920} \right)^{1/4}$$

Quantity	Value
$\text{Vol}(D_{IV}^5)$	$\pi^5/1920 = 0.15938525\dots$
α_{Wyler}	0.00729735...
$1/\alpha_{\text{Wyler}}$	137.036082
$1/\alpha_{\text{observed}}$	137.035999 (CODATA 2018)
Agreement	0.0001%

This is a purely geometric computation from the D_{IV}^5 volume. BST provides the physical reason for using this domain (Section 5.1–5.2 of WorkingPaper.md): D_{IV}^5 is the configuration space of the contact graph.

3. What the Casimir Stability Conjecture Actually Requires

The Casimir Stability Conjecture says $\rho = 137$ is a stable equilibrium — not that it is derived from the Casimir energy alone. This is a stronger claim:

1. Derivation of $\rho = 137$: Wyler formula (geometric, confirmed).
2. Stability of $\rho = 137$: Casimir stability conjecture (dynamical, open).

For the stability computation, the missing ingredient is the Bergman measure of D_{IV}^5 . The Bergman metric of the domain is not the flat product metric on $S^4 \times S^1$ — it includes non-trivial curvature contributions that couple R_s and R_b . The Casimir energy weighted by the Bergman measure would produce a ρ -dependent effective potential whose minimum could be at $\rho = 137$.

The three contributions needed: - S^4 bulk Casimir: ρ -independent at low T (confirmed) - S^1 winding Casimir: exponentially small for $\rho \gg 1$ (proved here) - Bergman boundary term: D_{IV}^5 curvature contributes a ρ -dependent piece — not yet computed

4. The S^4 Spectral Zeta at Key Values

Computed using Hurwitz zeta / Bernoulli polynomial method:

$\zeta_{S^4}(s)$	Value
$s = -2$	-0.040082
$s = -1$	+0.015170
$s = 0$	-1.005903
$s = 3$	+0.104816 (direct sum)

The regularized sum $\zeta_{S^4}(-1) = +0.015170$ determines the UV Casimir coefficient. Its sign (> 0) means the UV piece pushes ζ^{ren} upward with ρ , preventing equilibrium from this mechanism.

5. What Remains

Problem	Status	What's Needed
$\alpha = 1/137$ from Wyler	Confirmed	—
$\zeta_{S^4}(-1)$ via Hurwitz	Confirmed	—

Problem	Status	What's Needed
Winding modes negligible for $\rho \gg 1$	Proved Computed, corrected 2026-10-07 (Elie 5862: with the zero mode the winding piece is $-1/(6\rho)$, $\sim 0.3\%$ at 137, not 10^{-748} ; negligible for the minimum question)	—
Casimir min at $\rho = 137$ from UV+winding	Ruled out	—
Casimir stability from Bergman weighting	Superseded	See Section 6 below
Physical units R_b, R_s	Open	Follows from Wyler ratio

6. Reinterpretation: The Monotone Casimir Is the Expected Result

The monotone Casimir is not a failure — it is a confirmation that the stability mechanism is topological, not dynamical.

The argument. D_{IV}^5 is the unique bounded symmetric domain of Cartan type IV in 5 complex dimensions, determined by the BST contact structure with CR dimension $N_c + N_w = 5$. The Cartan classification is discrete. There is no continuous family of domains parameterized by ρ between which the system could slide. To change ρ would require changing the domain type, which requires changing the CR dimension, which requires changing the gauge structure of the Standard Model — a discrete jump, not a continuous deformation.

Therefore $\alpha = 1/137.036$ (the Wyler value for D_{IV}^5) is not the minimum of any potential. It is a geometric invariant of the domain — as fixed as π is for a circle. The Casimir energy has no extremum at $\rho = 137$ because the Casimir energy is irrelevant to the stability question. Stability is topological: no continuous path exists to a configuration with different α .

Why the Casimir computation was worth doing. The negative result rules out the simplest stability mechanism definitively and forced the correct understanding. A Casimir minimum would have been a weaker result: energy minima can be tunneled through if the barrier is finite. Topological rigidity cannot be violated by tunneling because there is no path to tunnel along.

What this changes in the framework. The Casimir Stability Conjecture (WorkingPaper.md Section 5.3) is superseded by the Topological Rigidity argument. The Wyler formula gives the *value* of α ; the Cartan classification gives the *stability*. No Casimir calculation is needed for either.

Research note, March 2026. Casey Koons. Code: `notes/bst_casimir_seeley_dewitt.py`. Data: `BST_Casimir_SeeleyDeWitt.csv`.

AI assistance: Claude Sonnet 4.6 (Anthropic) contributed to derivations, computations, and manuscript development.

Updated March 2026 to reflect topological rigidity reinterpretation.